

SLIDE 1: FIELD ORIGIN

Layout & Visuals

- **Left 55%:** Real documentary photograph of a Bangladeshi smallholder farmer inspecting potato/rice leaves with a smartphone in a rural field setting.
- **Right 45%:** Three stacked problem discovery cards on a soft parchment background.
- **Bottom:** Full-width anchor statement.

Slide Content

- **Header / Eyebrow:** THE STARTING POINT
- **Headline:** We started with farmers, not with a model.
- **Core Cards:**
 - **FIELD REALITY:** Agricultural decisions in Bangladesh are made under incomplete information, fluctuating weather, and severe time pressure.
 - **LANGUAGE GAP:** Farmers speak in regional dialects, colloquial terms, and mixed Banglish—not standardized textbook queries.
 - **ACCESS & CONNECTIVITY:** Expert extension officers are scarce (1:1,500 farmer ratio), and rural broadband is intermittent.
- **Anchor Statement:** *"The question was not 'How do we add AI to agriculture?' It was 'How can technology become genuinely useful and safe for a farmer?'"*

SLIDE 2: THE ADVISORY GAP & FAILURE MODES

Layout & Visuals

- **Structure:** 3-Way Problem Evolution Diagram on Left $\rightarrow$ Single Integrated Solution on Right.
- **Visual Style:** Minimalist high-contrast card comparison with clean iconography.

Slide Content

- **Header / Eyebrow:** THE ROOT PROBLEM
- **Headline:** The problem is not a lack of AI. It is a lack of accountable agricultural AI.
- **Comparison Columns:**
- **Generic LLMs:** Fluent & conversational, but hallucinates chemical dosages, invents unapproved pesticides, and lacks Bangladesh regulatory grounding.
- **Vanilla RAG:** Better text retrieval, but breaks under dialect variations, cannot handle multimodal conflict, and misbinds chemical attributes.
- **Vision-Only Tools:** Classifies leaf disease, but cannot provide holistic IPM guidance, weather-aware timing, or safe dosage calculations.
- **The Breakthrough (Right Card):**
- **KrishokChat: Language + Vision + Evidence + Safety + Rural Delivery** in one unified fail-closed architecture.
- **Footer:** *Our published research analyzed these exact failure modes before designing the platform architecture.*

SLIDE 3: RESEARCH-TO-PRODUCT TIMELINE

Layout & Visuals

- **Structure:** Horizontal 5-Stage Innovation Pipeline from Field Observation to System Deployment.
- **Accent:** Highlight "Research findings became engineering decisions".

Slide Content

- **Header / Eyebrow:** SCIENTIFIC FOUNDATION
- **Headline:** We learned from failure before we built the product.
- **Horizontal Pipeline:** [1. Field Discovery] → [2. Benchmark Corpus] → [3. Failure Analysis] → [4. Five-Tier Engine] → [5. Production Deployment] Farmer interviews 2,946 DAE/BARI docs Dialect & hallucination Deterministic + RAG PWA, SMS & Extension APIs
- **Three Key Milestones:**
 - **01 — Knowledge Foundation:** Built the first citation-grounded Bengali agricultural knowledge graph with 2,882 nodes and 11-slot contracts.
 - **02 — Retrieval Research:** Measured and quantified the register-gap: farmer dialect drops standard dense retrieval recall by up to 48%.
 - **03 — Safety Redesign:** Replaced unconstrained generation with a 5-Tier Resolution Ladder and relational dosage verification.
- **Key Takeaway:** Research findings directly shaped production architecture.

SLIDE 4: THE SYSTEM ARCHITECTURE

Layout & Visuals

- **Structure:** Clean multi-agent architectural flow diagram showing the 5-Tier Resolution Ladder and fail-closed pathways.
- **Hero Badge:** "No Evidence $\rightarrow$ No Unsafe Advice"

Slide Content

- **Header / Eyebrow:** MULTI-AGENT ADVISORY PIPELINE
- **Headline:** One advisory system, built around the farmer's real workflow.
- **Interactive Architecture Flow:**
- **Input Layer:** Multi-dialect Bengali Text, Voice Audio, Crop Photos.
- **Tier 0 — Deterministic Safety Gate:** Intercepts banned agrochemicals, crisis emergencies, and jailbreaks in **<0.1 ms** $\rightarrow$ Instant **16123** referral.
- **Tier 1 & 2 — Structured Fact-Base:** Resolves verified pathology facts deterministically in **3.8 ms** at **\$0.00 cost**.
- **Tier 3 — Hybrid RAG & Gemma-4:** Multi-source dense+sparse retrieval over 2,946 institutional documents.
- **Tier 4 — Relational Verifier:** Schema enforcement over 11 critical agronomic slots (dose bounds, safety gear, pre-harvest interval).
- **Delivery Channels:** Web PWA, Local Offline Cache, SMS Compression Engine.
- **Core Principle:** KrishokChat is not a chatbot wrapper. It is an accountable agricultural intelligence engine.

SLIDE 5: EMPIRICAL PROOF & BENCHMARKS

Layout & Visuals

- **Top Grid:** 4 Massive Hero Stat Callouts.
- **Bottom Grid:** Supporting empirical validation metrics from 36 completed experimental layers.

Slide Content

- **Header / Eyebrow:** RIGOROUS VALIDATION
- **Headline:** This is already far more than a prototype.
- **Hero Numbers:**
 - **2,946** Institutional Manuals Indexed (DAE, BARI, BRRI, BARC)
 - **97.0%** Certified Advisory Correctness (vs. 58–74% baseline LLMs)
 - **0.0%** Observed Critical Unsafe Acceptance (100% fail-closed safety)
 - **708×** Latency Speedup via Fact-Base (3.8 ms vs. 2,971 ms cloud generation)
- **Supporting Verification Grid:**
 - **11,000 / 11,000** Metamorphic Corruption Mutations Rejected (100.0% accuracy)
 - **0 / 1,400** Adversarial Jailbreak & Injection Attacks Accepted
 - **200 Cases** Double-Blind Evaluated by Certified Agronomists (4.82 / 5.0 Quality, 96.5% Approval)
 - **559 / 559** Backend Tests Passed • **50/50** Golden Replay Invariants Green

SLIDE 6: ECOSYSTEM IMPACT & COMMERCIAL SCALE

Layout & Visuals

- **Structure:** 3-Pillar Stakeholder Ecosystem (Farmers, Institutions, Agribusiness).
- **Flow:** Farmer Beneficiary $\rightarrow$ KrishokChat Platform $\rightarrow$ Institutional Partners $\rightarrow$ Sustainable Revenue.

Slide Content

- **Header / Eyebrow:** SUSTAINABLE VALUE CREATION
- **Headline:** Built for farmers. Scalable through institutions.
- **Three-Pillar Scale Model:**
- **For Farmers (Free / Direct Access):** Natural Bengali interaction, instant disease photo diagnosis, offline PWA access, zero-cost SMS advice.
- **For Extension Services & NGOs (B2G / NGO Partner):** Copilot for 14,000+ Sub-Assistant Agriculture Officers (SAAOs), case escalation management, regional disease outbreak surveillance.
- **For Agribusiness & Telecom (B2B API & Enterprise):** Certified agronomic API access, custom private knowledge packs, compliant agrochemical advisory integration.
- **Commercialization Philosophy:** *"The farmer is the primary beneficiary; institutions and enterprise provide the sustainable path to scale."*

SLIDE 7: RESPONSIBILITY, INCLUSION & SDG ALIGNMENT

Layout & Visuals

- **Structure:** 4 Clean Quadrants mapping Technology $\rightarrow$ Social & Environmental Impact $\rightarrow$ UN Sustainable Development Goals.

Slide Content

- **Header / Eyebrow:** ETHICAL & RESPONSIBLE AI
- **Headline:** Useful AI must also be responsible AI.
- **Four Impact Dimensions:**
 - **Safety & Poisoning Prevention:** Hard fail-closed guards prevent pesticide poisoning and misuse; automated routing to National Call Center 16123 and 999.
 - **Linguistic & Digital Inclusion:** Equal accuracy across 5 regional dialects and Banglish; offline caching bridges the rural 2G/3G connectivity gap.
 - **Environmental IPM & Soil Health:** Integrated Pest Management (IPM) prioritized first, driving a **77.1% reduction in unnecessary chemical usage**.
 - **Global & National SDG Alignment:**
 - **SDG 2 (Zero Hunger):** Increasing smallholder crop productivity and reducing disease loss.
 - **SDG 9 (Industry & Innovation):** Indigenous AI research deployed for national digital infrastructure.
 - **SDG 12 (Responsible Consumption):** Preventing chemical runoff and pesticide overdose.
 - **SDG 13 (Climate Action):** Weather-adaptive irrigation and disease risk early warnings.

SLIDE 8: THE ROADMAP & THE ASK

Layout & Visuals

- **Structure:** Two-Column Comparison (Current Accomplishment vs. Next Milestone) + Prominent Live QR Code and Links.

Slide Content

- **Header / Eyebrow:** DEPLOYMENT ROADMAP
- **Headline:** The technology exists. The next step is field deployment.
- **Two Column Matrix:**
- **What We Have Built (Today):**
 - Full-stack operational platform (Next.js + FastAPI)
 - 5-Tier safety and resolution pipeline
 - 2,946-document institutional knowledge index
 - Verified 97.0% accuracy across 36 empirical experiments
 - Full open test suite and live web demonstration
- **What Support & Funding Unlocks (Next 6–12 Months):**
 - Multi-district field pilot with 1,000+ active farmers in Bogura & Rangpur
 - Co-deployment pilot with DAE / SAAO extension officers
 - Expansion of on-device INT8 models for rice, corn, and wheat
 - Real-time soil sensor & satellite weather API integration
- **Call to Action:** *"Support the transition from a scientifically validated AI system to a national agricultural service."*
- **Access Links:**

- **Live Demo:** <https://krishokchat.vercel.app> (or local demo URL)
- **Research Dossier & Code:** GitHub / KrishokChat Repository